CAP 3330. Fall 2026

Assignment 2 (100 points)

Instructions

Create an R notebook to answer the questions presented below.

Question 1 (60 points)

Four different designs for a digital computer circuit are being studied to compare the amount of noise present. The following data have been obtained:

Circuit Design	**Noise Observed**				
Design 1	21	20	29	30	24
Design 2	30	31	33	26	30
Design 3	27	26	25	35	39
Design 4	35	41	33	28	38

Enter this data in R by following these steps:

- Create a vector called “design” by using this code:

design <- rep (c("1","2","3","4"), each= 5).

- Create a vector called “noise” and just enter all the values from the table. Enter the values by row (i.e., all the values from row 1 followed by all the values from row 2,...)
- Create a data frame titled “design_noise_df” with two columns (one column with each of the vectors you just created).

The goal is to run an ANOVA to find out if circuit design is related to the amount of observed noise.

a) (20 points) Run an exploratory analysis relevant to this problem. The analysis must include computing summary stats and doing a **dot plot**. Also, and most importantly, interpret the results of the exploratory analysis.

Requirement: You MUST do the exact same exploratory analysis we did in class. If you do something different (e.g., use different R functions, compute different summary stats, do different plots, etc than the ones we used in class), you will receive ZERO points for this part with the following comment: Intentionally failed to follow the instructions.

b) (5 points) State the hypotheses relevant to this problem.

c) (20 points) Run ANOVA and make the appropriate conclusion.

d) (15 points) Conduct a post-hoc test and discuss which designs lead to different average noise levels.

Question 2 (40 points)

20 pigs are assigned at random among 4 experimental groups. Each group is fed a different diet. The data are the pig's weight, in kilograms, after being raised on these diets for 10 months. We wish to determine whether the mean pig weights are the same for all 4 diets or not.

Download the CSV file “Pigs_Weights” from Canvas/ Module 3/ Files. Then, read it into R.

a) (5 points) Identify:
The dependent variable.
The factor.
The factor levels.

b) (5 points) State the hypotheses (Ho and Ha).

c) (10 points) Compute the average weight for each diet. Based on these means, can you conclude that diet is associated with a real difference in pig weight? Explain why or why not.

d) (20 points) Run ANOVA and make the appropriate conclusion. Use a significance level of 0.05.